Assignment 1　　杜乘旋　　12209095

1. (a) $2x_1+x_2=-2$, $4x_1-3x_2=6$　　$x_1=0$, $x_2=-2$　　$\therefore (0,-2)$

(b) $4x_1+3x_2=4$, $\frac{1}{3}x_1+4x_2=3$　　$21x_2=14$, $x_2=\frac{2}{3}$　　$\therefore x_1=\frac{1}{2}$　　$\therefore (\frac{1}{2},\frac{2}{3})$

(c) $x_2+x_3+x_4=0$, $3x_1+3x_3-4x_4=7$, $x_1+x_2+x_3+2x_4=6$, $2x_1+3x_2+x_3+3x_4=6$

$x_1+x_4=6$, $x_1+x_2+x_4=3$, $\therefore x_2=-3$, $x_3+x_4=3$

$3(6-x_4)+3(3-x_4)-4x_4=7$, $x_4=2$, $x_3=1$, $x_1=4$

$\therefore (4,-3,1,2)$

2. $5x_1+3x_2+x_3=3$, $2x_1+3x_2-4x_3=0$

3. $a_{11}x_1+a_{12}x_2=0$, $a_{21}x_1+a_{22}x_2=0$

$a_{11}x_1+a_{12}x_2=a_{21}x_1+a_{22}x_2$

$(a_{11}-a_{21})x_1=(a_{22}-a_{12})x_2$

If $a_{11}\neq a_{21}$, $x_1=\frac{a_{22}-a_{12}}{a_{11}-a_{21}}x_2$

$\therefore$ must be consistent

If $a_{11}=a_{21}$ and $a_{22}=a_{12}$, $x_1=x_2$

If $a_{11}=a_{21}$ and $a_{22}\neq a_{12}$, $x_2=0$, $x_1=$ all real number

4. (a) If $m_1\neq m_2$, $x_2=b_1+m_1x_1=b_2+m_2x$

$\therefore x_1=\frac{b_1-b_2}{m_2-m_1}$　　$x_2=\frac{m_2b_1-m_1b_2}{m_2-m_1}$

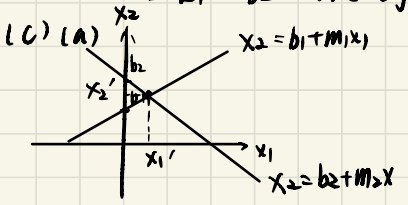
$\therefore$ will have unique solution

(b) If $m_1=m_2$, $b_2-b_1=x_1(m_1-m_2)$

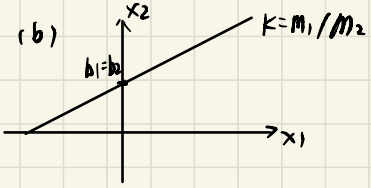
If $b_1\neq b_2$ left hand $\neq$ right hand

$\therefore b_1=b_2$, the system will be consistent

(c) (a)



(b)



5. row echelon form: (a) (c) (d) (e) (g) (h)

reduced echelon form: (a) (c) (d) (g)

6. (a) $\left[\begin{array}{cc|c} 1 & -2 & 3 \\ 2 & -1 & 9 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & -2 & 3 \\ 0 & 3 & 3 \end{array}\right]$ $\therefore x_2 = 1$, $x_1 = 5$ $(5, 1)$

(b) $\left[\begin{array}{ccc|c} 1 & -2 & 2 & 1 \\ 3 & 2 & -1 & 4 \\ 11 & 2 & 1 & 14 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -2 & 2 & 1 \\ 0 & 8 & -7 & 1 \\ 0 & 24 & -21 & 3 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -2 & 2 & 1 \\ 0 & 8 & -7 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$ $x_1 - 2x_2 + 2x_3 = 1$, $8x_2 - 7x_3 = 1$ $\therefore x_1 = \frac{5 - x_3}{4}$, $x_2 = \frac{1 + 7x_3}{8}$

$\left\{\left(\frac{5-\alpha}{4}, \frac{1+7\alpha}{8}, \alpha\right) \middle| \alpha \text{ real}\right\}$

(c) $\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 2 & 3 & -1 & -1 & 2 \\ 3 & 2 & 1 & 1 & 5 \\ 3 & 6 & -1 & -1 & 4 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & -3 & -3 & 2 \\ 0 & -1 & -2 & -2 & 5 \\ 0 & 3 & -4 & -4 & 4 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & -3 & -3 & 2 \\ 0 & 0 & -5 & -5 & 7 \\ 0 & 0 & 5 & 5 & -2 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & -3 & -3 & 2 \\ 0 & 0 & -5 & -5 & 7 \\ 0 & 0 & 0 & 0 & 5 \end{array}\right]$ $\therefore$ inconsistent

7. (a) Inconsistent

(b) $\left[\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right]$ $(4, -1)$

(c) $\left[\begin{array}{ccc|c} 1 & -2 & 4 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -2 & 0 & -11 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$ $x_1 - 2x_2 = -11$, $x_3 = 3$ $\therefore$ consistent, Infinitely many solutions

(d) $\left[\begin{array}{ccc|c} 1 & -2 & 2 & -2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right]$ $(4, 5, 2)$

(e) Inconsistent

8. $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ -1 & 4 & 3 & 2 \\ 2 & -2 & 0 & 3 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 6 & 4 & 3 \\ 0 & -6 & a-2 & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 6 & 4 & 3 \\ 0 & 0 & a+2 & 4 \end{array}\right]$ $\therefore a \neq -2$

9. (a) $(-2, 5, 3)$

(b) $\{(2+3\alpha, \alpha, -2) \mid \alpha \text{ real}\}$

(c) $\{(3-5\alpha+2\beta, \alpha, \beta, 6) \mid \alpha, \beta \text{ real}\}$;

10. (a) $\left[\begin{array}{cccc|c} 1 & 3 & 1 & 1 & 3 \\ 2 & -2 & 1 & 2 & 8 \\ 3 & 1 & 2 & -1 & -1 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 3 & 1 & 1 & 3 \\ 0 & -8 & -1 & 0 & 2 \\ 0 & -8 & -1 & -4 & -10 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 3 & 1 & 1 & 3 \\ 0 & -8 & -1 & 0 & 2 \\ 0 & 0 & 0 & -4 & -12 \end{array}\right]$ $x_4 = 3$, $8x_2 + x_3 = -2$, $x_1 + 3x_2 + x_3 = 0$ $\therefore \left\{\left(\frac{3}{4} - \frac{5\alpha}{8}, \frac{-2-\alpha}{8}, \alpha, 3\right) \middle| \alpha \text{ is real}\right\}$;

(b) $\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 2 & 1 & -1 & 3 & 0 \\ 1 & -2 & 1 & 1 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & -3 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & -3 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{array}\right]$ $x_1 + x_3 + x_4 = 0$, $-3x_3 + x_4 = 0$, $x_2 = 0$ $\therefore \left\{\left(-\frac{1}{4}\alpha, 0, \alpha, 3\alpha\right) \middle| \alpha \text{ is real}\right\}$

11. (a) coefficient matrix: $\begin{bmatrix} 4 & 5 & 3 & 3 & 4 \\ 2 & 3 & 1 & 0 & 1 \\ 3 & 4 & 2 & 1 & 1 \end{bmatrix}$ augmented matrix $\left[\begin{array}{ccccc|c} 4 & 5 & 3 & 3 & 4 & -5 \\ 2 & 3 & 1 & 0 & 1 & -3 \\ 3 & 4 & 2 & 1 & 1 & -1 \end{array}\right]$

(b) $\left[\begin{array}{ccccc|c} 1 & \frac{5}{4} & \frac{3}{4} & \frac{3}{4} & 1 & -\frac{5}{4} \\ 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{3}{2} & -1 & -\frac{1}{2} \\ 0 & \frac{1}{4} & -\frac{1}{4} & -\frac{5}{4} & -2 & \frac{11}{4} \end{array}\right] \to \left[\begin{array}{ccccc|c} 1 & \frac{5}{4} & \frac{3}{4} & \frac{3}{4} & 1 & -\frac{5}{4} \\ 0 & \frac{1}{4} & -\frac{1}{4} & -\frac{3}{4} & -\frac{1}{2} & -\frac{1}{4} \\ 0 & 0 & 0 & -\frac{1}{2} & -\frac{3}{2} & 3 \end{array}\right] \to \left[\begin{array}{ccccc|c} 2 & 3 & 1 & 0 & 1 & -3 \\ 0 & 1 & -1 & -3 & -2 & -1 \\ 0 & 0 & 0 & 1 & -3 & 6 \end{array}\right]$

$x_4 + 3x_5 = -6$ $\quad x_4 = -6 - 3x_5$

$x_2 - x_3 - 3x_4 - 2x_5 = -1$ $\quad x_2 - x_3 - 3(-6-3x_5) - 2x_5 = -1$

$2x_1 + 3x_2 + x_3 + x_5 = -3$ $\quad x_2 - x_3 + 18 + 7x_5 = -1$

$x_1 = 27 + 10x_5 - 2x_3$

$x_2 = x_3 - 19 - 7x_5$

$\therefore \{(27 + 10\beta - 2\alpha, \alpha - 19 - 7\beta, \alpha, -6 - 3\beta, \beta) \mid \alpha, \beta \text{ is real}\}$

12. Assume matrix A reduces to two RREF R and S, and $R \neq S$

$R = \left[\begin{array}{c|c} I_n & r \\ \hline 0 & 0 \end{array}\right]$ or $\left[\begin{array}{c|c} I_n & 0 \\ \hline 0 & \vdots \end{array}\right]$, $S = \left[\begin{array}{c|c} I_n & s \\ \hline 0 & 0 \end{array}\right]$ or $\left[\begin{array}{c|c} I_n & 0 \\ \hline 0 & \vdots \end{array}\right]$

If follows that R and S are row equivalent since selection of columns does not affect row equivalence and that they reduced but not equal.

R has a unique solution r or is insistent. S has a unique solution s or is inconsistent since the systems are equivalent. $r = s$ or both are inconsistent

Either way $R = S$. a contradiction

$\therefore$ Is unique

13. $A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$

$AB = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1-1 & 1-1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

14. (a) $\left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & a-3 & b-2 \end{array}\right]$ $a = 5$, $b = 4$ have Infinitely solutions

(b) $\to \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & a-5 & b-4 \end{array}\right]$ Inconsistent: $a = 5$, $b \neq 4$

15. $A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$ $A^2 = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = A$

$A = A^2 = A^3 = A^n = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$

16. $A = \begin{bmatrix} -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ $A^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I$

$A^3 = A$

$\therefore A^{2n} = I \quad A^{2n+1} = A$